Design of the Numerical Experiments

A Signal–Strength Map $f(\text{signal}, \text{strength}) \to \boldsymbol{w}$
and the Five Potential Experiments

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Contents

Common Notations

We fix N risky assets. Bold lower-case letters denote vectors in $\mathbb{R}^N$ and $\mathbf{1} = (1, \dots, 1)^\top$. A *portfolio* is a weight vector $\boldsymbol{w} \in \mathbb{R}^N$ with $\mathbf{1}^\top \boldsymbol{w} = 1$; the long-only case adds $\boldsymbol{w}_i \geq \mathbf{0} \quad \forall i$, i.e. $\boldsymbol{w}$ lives on the simplex $\Delta^{N-1} = \{\boldsymbol{w} \geq \mathbf{0} : \mathbf{1}^\top \boldsymbol{w} = 1\}$.

Returns(used in Experiment 1). $\boldsymbol{R} = (R_1, \dots, R_N)^\top$ are one-period simple returns with mean $\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{R}]$ and covariance $\boldsymbol{\Sigma} = \mathrm{Cov}(\boldsymbol{R}) \in \mathbb{R}^{N \times N}$, assumed positive definite. We use the three Markowitz scalars

$$A = \mathbf{1}^\top \boldsymbol{\Sigma}^{-1} \mathbf{1}, \qquad B = \mathbf{1}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}, \qquad C = \boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}. \tag{1}$$

Signal and strength (used in Experiments 2). A *signal* is a cross-sectional score $\boldsymbol{s} = (s_1, \dots, s_N)^\top \in \mathbb{R}^N$, one number per name at a single instant. Where the signal is built from firm characteristics $\hat{\boldsymbol{x}}_i \in \mathbb{R}^K$ we write $s_i = \boldsymbol{\theta}^\top \hat{\boldsymbol{x}}_i$: here $\boldsymbol{\theta} \in \mathbb{R}^K$ is the *direction* (which characteristics matter and how much they are trusted *relative to one another*) and the scalar $\beta \geq 0$ is the *strength* (how hard the whole signal is acted upon). These two roles are kept strictly separate throughout: $\boldsymbol{\theta}$ chooses the ranking, β chooses the concentration along that fixed ranking.

The map(used in Experiment 2,3,4,5). The object under study is

$$f : (\boldsymbol{s}, \beta) \longmapsto \boldsymbol{w}(\boldsymbol{s}, \beta) \in \Delta^{N-1}, \qquad \text{with} \qquad \begin{cases} \boldsymbol{w}(\boldsymbol{s}, 0) = \frac{1}{N}\mathbf{1} & \text{(no information)}, \\ \boldsymbol{w}(\boldsymbol{s}, \beta) \to \boldsymbol{e}_{i^\star} & \text{as } \beta \to \infty, \text{ Oracle} \end{cases} \tag{2}$$

where $i^\star = \arg\max_i s_i$ and $\boldsymbol{e}_{i^\star}$ is the corresponding simplex vertex. The remainder of this note is a sequence of experiments that instantiate, derive, simulate, and stress-test (2).

Chapter 1

The No-Information Limit for $\beta = 0$

Research 1. At the $\beta \to 0$ end the signal carries no differentiating information. The experiment asks what the map *should* return there, and benchmarks the equal-weight portfolio against the mean–variance portfolio and against a naive–sophisticated combination.

1.1 Why mean–variance already assumes more than nothing

Mean–variance optimisation treats $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ as jointly sufficient for the return distribution. This is exact only when returns are elliptical (Gaussian being the canonical case) *or* when utility is quadratic, so that moments beyond the second never enter the objective. It is therefore strictly more information than the no-signal baseline and strictly less than oracle knowledge — it is one interior point of (2), not an endpoint.

1.2 The three candidate portfolios

Definition 1.1 (Candidates at low signal)**.**

(i) **Equal weight.** $\boldsymbol{w}_{\mathrm{eq}} = \frac{1}{N}\mathbf{1}$.

(ii) **Mean–variance (γ-problem).** With absolute risk aversion $\gamma > 0$, the unconstrained certainty-equivalent objective $\max_{\boldsymbol{w}}\ \boldsymbol{\mu}^\top \boldsymbol{w} - \frac{\gamma}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma}\boldsymbol{w}$ gives

$$\boldsymbol{w}^\star_{\mathrm{unc}} = \tfrac{1}{\gamma}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}, \tag{1.1}$$

and under the budget constraint $\mathbf{1}^\top \boldsymbol{w} = 1$,

$$\boldsymbol{w}^\star = \underbrace{\frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}}{A}}_{\boldsymbol{w}_{\mathrm{gmv}}} + \frac{1}{\gamma}\left(\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} - \frac{B}{A}\boldsymbol{\Sigma}^{-1}\mathbf{1}\right), \tag{1.2}$$

i.e. the global minimum-variance portfolio $\boldsymbol{w}_{\mathrm{gmv}}$ plus a self-financing tilt (the bracket sums to zero) scaled by $1/\gamma$.

(iii) **Naive combination** (inspired by Tu and Zhou's *Markowitz meets Talmud*, §2.2):

$$\hat{\boldsymbol{w}}_c = (1-\delta)\,\boldsymbol{w}_{\mathrm{eq}} + \delta\,\hat{\boldsymbol{w}}^\star, \qquad \delta^\star = \frac{\tau}{\tau+\lambda}, \tag{1.3}$$

where $\delta \in [0,1]$ is itself a strength-like mixing weight between the agnostic and the estimated-optimal portfolio.

Remark 1.2 (Two-fund separation). Every frontier (efficient) portfolio is an affine combination of the two fixed portfolios $\boldsymbol{g} = \boldsymbol{\Sigma}^{-1}\mathbf{1}/A$ and $\boldsymbol{d} = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}/B$, because the first-order condition on the budget hyperplane always reads $\boldsymbol{\Sigma}\boldsymbol{w} = (\text{scalar})\mathbf{1} + (\text{scalar})\boldsymbol{\mu}$. Equations (1.1)–(1.2) are the two members of this family selected by the risk-aversion γ.

Remark 1.3 (Shadow price of the return floor:Lagrange duality). Writing the target-return problem $\min_{\boldsymbol{w}} \frac{1}{2}\boldsymbol{w}^\top\boldsymbol{\Sigma}\boldsymbol{w}$ s.t. $\boldsymbol{\mu}^\top\boldsymbol{w} \geq \mu_{\min}$, $\mathbf{1}^\top\boldsymbol{w} = 1$ with multipliers η (floor) and λ (budget) gives $\boldsymbol{\Sigma}\boldsymbol{w} = \eta\boldsymbol{\mu} + \lambda\mathbf{1}$. Matching to the γ-problem identifies $\eta = 1/\gamma$: the shadow price of the return floor is the reciprocal of risk aversion. Tightening the floor (raising $\mu_{\min}$) is equivalent to lowering γ — less risk-averse, more greedy for return, higher volatility.

1.3 Experiment

Experiment 1.1 (The agnostic limit)**.** Compute $\mathbb{E}[\,u(\boldsymbol{w}^\top\boldsymbol{R})]$ for $\boldsymbol{w}_{\mathrm{eq}}$, $\boldsymbol{w}^\star(\gamma)$ and $\hat{\boldsymbol{w}}_c$ over repeated samples, and verify that as the signal degrades, equal weighting is the point the map should collapse to. Here u is the utility function; I take the CARA utility $u(x) = -e^{-\gamma x}$, where $\gamma > 0$ is the absolute risk-aversion parameter.
The expected empirical finding echoes DeMiguel–Garlappi–Uppal: at short estimation windows $\boldsymbol{w}_{\mathrm{eq}}$ is hard to beat out of sample, because the estimation error injected through $\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$ overwhelms the optimisation gain. This motivates a *finite* strength: the signal must temper, not drive, the raw optimisation.

Goal. Initially, I wanted to decide whether to use the Markowitz, the equal-weighted, or the combined portfolio at the lower end of the map. However, I noticed — and confirmed with Felix — that knowing the joint return distribution conflicts with the no-information assumption. Hence, in the no-information limit, it is not possible to construct the Markowitz portfolio, since that would require the first and second moments. Thus, at the lower limit, the only sensible comparison is between the equal-weighted portfolio and randomly picking a single name. The equal-weighted portfolio is "obviously" the better choice, which can be shown easily via either (1) entropy maximisation, or (2) the Lagrange multiplier method as in Proposition 2.1.

Chapter 2

The Gibbs–Softmax Map

Research 2. We construct the interior of the map from a single variational principle, obtain the softmax weight rule, and recover the linear parametric policy- the two boundary conditions as its small-strength expansion.

2.1 A maximum-entropy variational programme

Lead with the generic programme: among all long-only portfolios that achieve a prescribed exposure to the signal, choose the most diversified one, where diversification is measured by the Shannon entropy of the weights $\mathrm{H}(\boldsymbol{w}) = -\sum_i w_i \ln w_i$.

$$\max_{\boldsymbol{w}\in\Delta^{N-1}} \mathrm{H}(\boldsymbol{w}) \qquad \text{s.t.} \qquad \sum_{i=1}^{N} w_i s_i = \bar{S}, \quad \sum_{i=1}^{N} w_i = 1. \tag{2.1}$$

Proposition 2.1 (Softmax is the unique solution)**.** *The solution of* (2.1) *is the Gibbs (softmax) weight*

$$w_i(\boldsymbol{s}, \beta) = \frac{e^{\beta s_i}}{\sum_{j=1}^{N} e^{\beta s_j}}, \qquad i = 1, \ldots, N, \tag{2.2}$$

where β is the Lagrange multiplier on the signal-exposure constraint and plays the role of strength*.*

Proof. With multipliers λ (budget) and β (exposure), the Lagrangian is $\mathcal{L} = -\sum_i w_i \ln w_i + \lambda\big(\sum_i w_i - 1\big) + \beta\big(\sum_i w_i s_i - \bar{S}\big)$. Setting $\partial\mathcal{L}/\partial w_i = -\ln w_i - 1 + \lambda + \beta s_i = 0$ gives $w_i = \exp(\lambda - 1)\exp(\beta s_i)$; imposing $\sum_i w_i = 1$ eliminates λ and yields (2.2). Strict concavity of H on the simplex makes the stationary point the unique maximiser. Positivity $w_i > 0$ is automatic from the exponential — the entropy's log-barrier is exactly what makes the map intrinsically long-only. □

Remark 2.2 (Why symmetry alone is not enough)*.* Pure permutation symmetry of the N indistinguishable names does not by itself select $\frac{1}{N}\mathbf{1}$: it fixes all simplex vertices equally well too. It is the *strict concavity* of the entropy that uniquely chooses the uniform vector. This is the precise reason the $\beta \to 0$ boundary condition in (2) is equal weighting rather than some other symmetric object.

2.2 The two limits

Proposition 2.3 (Boundary behaviour). *Let $i^\star = \arg\max_i s_i$ be unique. Then $\lim_{\beta\to 0} w_i(\boldsymbol{s}, \beta) = \frac{1}{N}$ for every i, and $\lim_{\beta\to\infty} w_i(\boldsymbol{s}, \beta) = \begin{cases} 1 & \text{if } i = i^\star, \\ 0 & \text{otherwise.} \end{cases}$.*

Proof. As $\beta \to 0$, $e^{\beta s_i} = 1 + \beta s_i + O(\beta^2) \to 1$, so every ratio tends to $1/N$. As $\beta \to \infty$, dividing numerator and denominator by $e^{\beta s_{i^\star}}$ leaves 1 in the $i^\star$ slot and terms $e^{\beta(s_i - s_{i^\star})} \to 0$ elsewhere. □

Remark 2.4 (Strength rescales, it does not re-rank). Since β multiplies every score by the same positive constant, the ordering of $\boldsymbol{s}$ is invariant: if $s_i > s_j$ for one $\beta > 0$ then $s_i > s_j$ for all $\beta > 0$. Thus β moves the portfolio along a *fixed* ranking from the barycentre $\frac{1}{N}\mathbf{1}$ towards the vertex $\boldsymbol{e}_{i^\star}$, stretching or compressing the gaps between scores but never changing the direction. This is the concrete meaning of "strength controls concentration, not direction." Only the signal is determined, the weight rank is also deterministic.

2.3 The complete chain and the recovered linear policy

The score itself is supplied by the Brandt–Santa-Clara–Valkanov (BSCV) parametric policy, giving the chain

$$\underbrace{\hat{\boldsymbol{x}}_i}_{\text{characteristics}} \xrightarrow{\boldsymbol{\theta}} \underbrace{s_i = \boldsymbol{\theta}^\top \hat{\boldsymbol{x}}_i}_{\text{signal / score}} \xrightarrow{\beta} \underbrace{w_i}_{\text{weight}}.$$

BSCV fit $\boldsymbol{\theta}$ once, across all names and dates, by maximising realised average utility, $\max_{\boldsymbol{\theta}} \frac{1}{T}\sum_t u\big(\sum_i f(\hat{\boldsymbol{x}}_{i,t}; \boldsymbol{\theta}) R_{i,t+1}\big)$, and use the linear policy $w_{i,t} = \bar{w}_{i,t} + \frac{1}{N_t}\boldsymbol{\theta}^\top \hat{\boldsymbol{x}}_{i,t}$.

Proposition 2.5 (BSCV is the small-strength limit of softmax via Taylor Expansion). *If the signal is standardised to zero cross-sectional mean ($\bar{s} = \frac{1}{N}\sum_j s_j = 0$, as BSCV assume after standardisation), then to first order in β*

$$w_i(\boldsymbol{s}, \beta) = \frac{1}{N} + \frac{\beta}{N} s_i + O(\beta^2). \tag{2.3}$$

Proof. Expand $e^{\beta s_i} \approx 1 + \beta s_i$ and $\sum_j e^{\beta s_j} \approx N(1 + \beta \bar{s})$, so $w_i \approx \frac{1}{N}(1 + \beta s_i)(1 - \beta \bar{s}) = \frac{1}{N} + \frac{\beta}{N}(s_i - \bar{s}) + O(\beta^2)$. With $\bar{s} = 0$ this is (2.3). □

Hence the BSCV linear tilt is not an independent model but the $\beta \to 0$ Taylor expansion of the softmax spine — exactly the "recovered special case" the framework is required to produce.

2.4 Numerical illustration

For three stocks with $\boldsymbol{s} = (2, 1, 0)^\top$, formula (2.2) gives Table 2.1: the mass migrates smoothly from the barycentre towards the leading name as β grows. Note, here the weight ranks never change.

β	w_1	w_2	w_3
0	0.333	0.333	0.333
1	0.665	0.245	0.090
3	0.950	0.047	0.002

Table 2.1: Softmax weights for $\boldsymbol{s} = (2, 1, 0)^\top$ at three strengths.

Experiment 2.1 (Signal-to-weight map)**.** Take the signal as given (the project's simplification relative to BSCV, which must first *estimate* $\boldsymbol{\theta}$) and demonstrate that $\boldsymbol{s} \mapsto \boldsymbol{w}$ is fully determined by (2.2), tracing the trajectory from $\frac{1}{N}\mathbf{1}$ to $\boldsymbol{e}_{i^\star}$ as β sweeps from 0 to large.

Goal. We have already proved that this softmax map satisfies the two boundary conditions, though its limitation is that it assumes a long-only portfolio. The next goal is to see whether we can link this map to the Markowitz portfolio: that is, whether there exists a specific β that produces a map similar to the Markowitz portfolio, and whether we can generalise the framework from there.

Chapter 3

Signals in a Geometric Brownian Motion

Research 3. We drive the map with simulated prices. The central design question is how to inject a signal that is informative about the future without secretly *being* the future — the clean-versus-leakage line. The key here is that we still have a diffusion term which is generated from a Gaussian distribution and can't be predicted even though we incorporate the drift (and some noise) in our signal.

3.1 One-step price and return, just the Monte Carlo simulation from GBM

Each stock follows geometric Brownian motion $\mathrm{d}S_i = \mu_i S_i \,\mathrm{d}t + \sigma_i S_i \,\mathrm{d}W_i$. Passing to logs turns this into arithmetic Brownian motion, and over a single step of length Δt the simple return is

$$R_i = \exp\Big[\big(\mu_i - \tfrac{1}{2}\sigma_i^2\big)\Delta t + \sigma_i\sqrt{\Delta t}\,\phi_i\Big] - 1, \qquad \phi_i \sim \mathcal{N}(0,1). \tag{3.1}$$

3.2 The clean signal

Definition 3.1 (Drift-tracking signal)**.** The signal tracks the *drift*, not the realised return:

$$s_i = \mu_i + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma_\varepsilon^2), \qquad \varepsilon_i \perp \phi_i. \tag{3.2}$$

Each stock thus carries exactly two independent random variables: the signal noise ε_i, drawn at $t = 0$ when the view is formed, and the diffusion shock ϕ_i, which realises the price. Because $\varepsilon_i \perp \phi_i$, the quantity that decides the future (ϕ_i) is never leaked into the signal.

Remark 3.2 (Why leakage matters). If instead one set $s_i = R_i$ (signal = realised return), then $\arg\max_i s_i$ would be the best performer *by construction* and the softmax would merely concentrate on a known winner: "signal" would be identical to "knowing the outcome," and any apparent out-performance would be spurious. Equation (3.2) is the minimal device that keeps the clean-versus-cheating line explicit. Conversely, the oracle sweep uses $s_i = \mu_i$ (i.e. $\sigma_\varepsilon \to 0$): perfect knowledge of the drift, though the diffusion shock remains.

3.3 Monte-Carlo algorithm

For a chosen utility u (default CARA utility function, $u(x) = -e^{-\gamma x}$):

1. draw drifts μ_i;
2. form signals $s_i = \mu_i + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma_\varepsilon^2)$;
3. allocate $w_i = e^{\beta s_i} / \sum_j e^{\beta s_j}$;
4. draw diffusion shocks ϕ_i, obtain $\boldsymbol{R}$ via (3.1), and score $u(\boldsymbol{w}^\top \boldsymbol{R})$;
5. average over many draws of $(\boldsymbol{\varepsilon}, \boldsymbol{\phi})$ to estimate $\mathrm{EU}(\beta) = \mathbb{E}\, u(\boldsymbol{w}^\top \boldsymbol{R})$.

There is deliberately no path, no rebalancing and no compounding: a single step isolates the structural effect of strength. Only two noise levels enter — signal quality σ_ε and diffusion $\sigma_i \sqrt{\Delta t}$.

3.4 Experiments

Experiment 3.1 (Strength sweep)**.** Sweep β from 0 upward and plot $\mathrm{EU}(\beta)$ relative to the $\beta = 0$ (equal-weight) baseline. Test whether an interior maximiser $\beta^\star$ exists with $\mathrm{EU}(\beta^\star)$ largest.

Goal. I want to check whether $\mathrm{EU}(\beta)$ is monotonic — that is, whether the utility always increases with a stronger signal — or whether there is a limit on signal strength, beyond which diminishing utility kicks in. If such a limit exists, the interior maximiser $\beta^\star$ is the strength at which the gain from tilting on the signal is exactly balanced by the risk of concentrating too much.

Experiment 3.2 (Signal quality)**.** Vary σ_ε^2 (clean / medium / noisy) and track $\beta^\star$. Prediction: the noisier the signal, the smaller $\beta^\star$, tending to $\beta^\star \to 0$ as $\sigma_\varepsilon \to \infty$.

Goal. I want to understand the relationship between signal strength and signal noise. Specifically, whether signal noise severely affects the optimal strength or has only a small impact. This tells us how much we can trust the signal, and therefore how hard we should tilt on it, as the noise level varies.

Experiment 3.3 (Oracle versus noisy)**.** Overlay two sweeps (Table 3.1). The *vertical* gap between the curves is the cost of signal noise — the utility lost because the signal is corrupted rather than perfect. The *horizontal* gap explains why optimal strength retreats under noise: with a clean drift, a high signal really is a high drift and one can bet hard on a single name ($\beta^{\star}_{\text{oracle}}$ large); with noise a large $s_i = \mu_i + \varepsilon_i$ may be a lucky draw of ε_i rather than a genuinely high μ_i, so the optimal strength must be smaller than with a perfect reading of the drift. For these gaps, I have sketched a brieft diagram in my handwritten note.

Goal. I want to separate the two distinct ways that noise hurts the portfolio: how much utility is lost at a fixed strength (the vertical gap), and how much the optimal strength itself is pulled back (the horizontal gap). Isolating the two confirms that noise both lowers the achievable utility and forces a more cautious tilt, rather than these being a single effect.

Sweep	Signal for softmax	Interpretation
Oracle	$s_i = \mu_i$ (true drift)	best achievable, perfect drift
Noisy	$s_i = \mu_i + \varepsilon_i$	closer to reality, imperfect signal

Table 3.1: The two β-sweeps compared in Experiment 3.3.

Chapter 4

Comparing Weight Maps

Research 4. As Felix mentioned about a more generalised framework, i decided to test research 3 on the following 3 maps. Now the signal $\boldsymbol{s}$ and the strength grid are held fixed and the *map* itself is varied. Three candidates are placed on one axis.

4.1 The three maps

$$\text{Softmax (long-only, bounded):} \quad w_i = \frac{e^{\beta s_i}}{\sum_j e^{\beta s_j}}, \tag{4.1}$$

$$\text{BSCV linear (affine, unbounded, allows short):} \quad w_i = \frac{1}{N} + \frac{\beta}{N}\, s_i, \tag{4.2}$$

$$\text{BSCV long-only (truncated, not smooth):} \quad w_i^{+} = \frac{\max\bigl(0,\ \frac{1}{N} + \frac{\beta}{N} s_i\bigr)}{\sum_j \max\bigl(0,\ \frac{1}{N} + \frac{\beta}{N} s_j\bigr)}. \tag{4.3}$$

4.2 Agreement at low strength, divergence at high strength

Proposition 4.1 (Coincidence near $\beta = 0$)**.** *At $\beta = 0$ all three maps equal $\frac{1}{N}\mathbf{1}$. For standardised signals ($\bar{s} = 0$) the softmax and linear maps further agree to first order, both equalling $\frac{1}{N} + \frac{\beta}{N} s_i$ by (2.3). In this sense the linear map is the small-strength invariant of the softmax.*

At large β the three separate sharply, and the separation is geometric — *simplex versus hyperplane*:

- **Softmax** saturates: $\boldsymbol{w} \to \boldsymbol{e}_{i^\star}$, the curve $\mathrm{EU}(\beta)$ flattening onto the single-best-stock utility. The weights never leave the simplex.
- **Linear** runs off the simplex into unbounded long–short leverage; variance explodes and $\mathrm{EU}(\beta)$ turns down hard. It can afford only a little strength before leverage wrecks it, so its peak is early and low.

- **Truncation** sits between the two, with kinks where names hit zero and are clipped at particular values of β.

4.3 Experiment

Experiment 4.1 (Which map?). Plot the three EU(β) curves on one axis and compare their shape, boundedness and peak height $\beta^\star$. The optimal strength differs by map: softmax can run to much higher β before saturating, whereas the linear map peaks early. The purpose is to show that at high strength the choice of map genuinely matters — one cannot simply assume softmax — while recording that softmax is the unique candidate that stays on the simplex and inherits the maximum-entropy / Bera–Park exponential justification of Chapter 2.

Goal. So far I cannot think of a more general map that recovers all of the feasible maps above (those satisfying the two boundary conditions) as special cases. Hence I would like to know whether EU actually differs much when I switch from one map to another. If the difference is small, then it may be reasonable to simply commit to a single map; if it is large, the choice of map matters in its own right and cannot be treated as a modelling convenience.

Chapter 5

Multiple Signals and Vector Strength

Research 5. We relax the single-signal assumption. Each stock now carries K signals (e.g. momentum, value, trend), and we ask how the strength should act when the signals differ in quality. This may also be relevant to our Wednesday meeting on how to combine signals — for example, whether to simply take the numerical average of the signals, or to optimise the loadings $\boldsymbol{\theta}$ from some objective function.

5.1 Scalar strength: shared trust

Combining K standardised signals through the direction $\boldsymbol{\theta} \in \mathbb{R}^K$ gives the score and softmax

$$s_i = \boldsymbol{\theta}^\top \hat{\boldsymbol{x}}_i, \qquad w_i = \frac{e^{\beta s_i}}{\sum_j e^{\beta s_j}}. \tag{5.1}$$

Here $\boldsymbol{\theta}$ is the direction (how much each signal *type* is trusted relative to the others) and the scalar β is the magnitude (how hard the combined view is acted on). The coefficient on signal k is $\beta\theta_k$, shared across all names.

Remark 5.1 (Limitation of a scalar). A single $\beta \in \mathbb{R}$ forces every signal to be trusted in the *same* fixed proportion regardless of its individual noise. If one signal is clean and another is mostly noise, a scalar strength cannot tell them apart.

5.2 Vector strength: per-signal trust

Let the strength be a vector $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_K)^\top$ and fold it directly into the score:

$$s_i = \sum_{k=1}^{K} \beta_k \, \hat{x}_{ik} = \boldsymbol{\beta}^\top \hat{\boldsymbol{x}}_i, \qquad w_i = \frac{\exp\big(\sum_k \beta_k \hat{x}_{ik}\big)}{\sum_j \exp\big(\sum_k \beta_k \hat{x}_{jk}\big)}. \tag{5.2}$$

Because the strength is now inside the score, no separate shared scalar is needed. The coefficient on signal k is simply β_k, and the natural choice is precision weighting,

$$\beta_k \ \propto \ \frac{1}{\sigma^2_{\varepsilon,k}}, \tag{5.3}$$

so that each signal is trusted in inverse proportion to its noise, exactly as a Bayesian precision would dictate.

Remark 5.2 (Genuinely new content). The scalar case (5.1) is the special case $\boldsymbol{\beta} = \beta\,\boldsymbol{\theta}$ of (5.2). The vector case cannot be collapsed back to a single global scalar without discarding the per-signal information, which is precisely why the multi-signal extension adds structure rather than notation. All previous experiments treat β as a scalar; for per-signal strength, however, β must be a vector, which raises the dimension of the map and hence the difficulty of finding a generalised framework.

Experiment 5.1 (Vector strength)**.** Compare EU under the scalar rule (5.1) and the vector rule (5.2), driving each signal k with its own noise level $\sigma_{\varepsilon,k}$ in the GBM simulation of Chapter 3. Test whether the precision weighting (5.3) recovers, and improves on, the best scalar β, and whether the advantage grows as the signals become more heterogeneous in quality. Felix and I also discussed possibly adding a correlation/covariance matrix for the signals, so that we can drop the assumption that all the signals are independent.

Goal. We aim for a more general map by allowing multiple signals and dropping the independence assumption between them. The limitation, however, is that I may still need to use this specific softmax map for the experiment, rather than the more general map that I currently cannot think of.

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